Chapter 15

Cardiovascular, Respiratory, and Digestive Systems

1. The Cardiovascular and Lymphatic Systems
2. The Respiratory System
3. The Digestive System
4. The Excretory System

Lesson 1 — The Cardiovascular and Lymphatic Systems

Your heart is a fist-sized muscle with four chambers pumping blood through a network that, if laid end to end, would circle the Earth nearly two and a half times. Blood has four main components:

Plasma	~55% of blood volume; pale yellow liquid carrying nutrients, proteins, hormones, and salts
Red blood cells	~40% of blood; contain hemoglobin that carries oxygen and carbon dioxide
White blood cells	Immune defenders — engulf pathogens, produce antibodies, fight allergic reactions
Platelets	Tiny cell fragments that trigger blood clotting when a vessel is damaged

Blood Pressure — Updated Guidelines

Normal	Below 120/80
Elevated	120–129/less than 80
Stage 1 Hypertension	130–139/80–89 (new lower threshold from 2017 AHA guidelines — was 140/90)
Stage 2 Hypertension	140/90 or higher

Lesson 2 — The Respiratory System

Your respiratory system performs a vital two-way gas exchange — delivering oxygen to the blood and removing carbon dioxide. Air travels: Nose/mouth → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → **Alveoli** (tiny air sacs where gas exchange happens).

Asthma affects about 1 in 13 Americans — approximately **28 million** (nearly double older estimates). COVID-19 vaccines represent a major new respiratory disease prevention tool since this textbook was written.

Lesson 3 — The Digestive System

Food travels: Mouth (mastication + enzymes) → Esophagus (peristalsis) → Stomach (gastric juices + pepsin) → Small intestine (90% of absorption via villi) → Large intestine (water absorption) → Elimination.

Colorectal cancer is now the **#2 cause of cancer death** in the U.S. and has become the **leading cause in men under 50**. The American Cancer Society now recommends starting colorectal cancer screening at **age 45** (lowered from 50 in 2021).

Lesson 4 — The Excretory System

Excretion removes metabolic wastes that would become toxic if allowed to build up. The kidneys contain over a million **nephrons** — microscopic filtering units. The liver is also a powerful detoxifier, processing drugs and alcohol. Drink at least 8 cups of water daily to keep kidneys flushing properly.

Chapter Vocabulary

Lesson 1

Plasma	The liquid portion of blood (~55%), carrying nutrients, proteins, hormones, and salts
Hemoglobin	The oxygen-carrying protein in red blood cells, containing iron
Platelets	Blood cell fragments that initiate clotting when a vessel is damaged
Capillaries	The smallest blood vessels; the site of gas and nutrient exchange between blood and cells
Blood pressure	The force blood exerts against artery walls as the heart pumps

Lesson 2

Diaphragm	The dome-shaped muscle below the lungs that drives breathing
Trachea	The windpipe; carries air from the throat to the bronchi
Asthma	A chronic inflammatory condition in which airways narrow, causing difficulty breathing
Emphysema	A progressive, irreversible disease that destroys alveolar walls; almost always caused by smoking

Lesson 3

Mastication	The mechanical process of chewing
Peristalsis	Rhythmic, involuntary muscle contractions that move food through the digestive tract
Gastric juices	Stomach secretions containing hydrochloric acid and pepsin that chemically digest food
Appendicitis	Inflammation of the appendix; requires emergency surgery

Lesson 4

Nephrons	The microscopic filtering units of the kidney; each kidney has over a million
Ureters	Tubes connecting the kidneys to the bladder
Cystitis	Inflammation of the bladder, usually from bacterial infection
Hemodialysis	A medical procedure using a machine to filter waste from the blood when the kidneys fail